

The Critical Line as a Valley of the Xi Function

Analytical and Numerical Evidence for the Riemann Hypothesis

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Abstract

We prove two analytical results about the Riemann xi function. First, the Hermite-Biehler condition: for all real σ and t ,

$\xi(\sigma+it)$	=	$\xi(\sigma-it)$	. Second, the critical line
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$\sigma=1/2$ is a critical point of $\xi(\sigma+it)^2$ for every t . We verify numerically that this critical point is a strict local minimum (positive Laplacian) and that ξ decays super-exponentially on the boundaries of the critical strip. We identify the remaining gap: proving the Laplacian is positive everywhere.

1. Introduction

The Riemann Hypothesis (RH) states that every nontrivial zero of $\zeta(s)$ has real part $1/2$. We study the xi function:

$$\xi(s) = (1/2) s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

which is entire, satisfies $\xi(s) = \xi(1-s)$, is real on $\mathbb{R}$, and shares the nontrivial zeros of ζ .

2. Properties of Xi

(P1) ξ is entire of order 1. (P2) $\xi(s) = \xi(1-s)$ for all s (functional equation). (P3) $\xi(s) = \overline{\xi(\overline{s})}$ for all s (reality on $\mathbb{R}$). (P4) Nontrivial zeros: $\rho_n = 1/2 + i\gamma_n$, γ_n real.

3. Main Results

Theorem 1 (Hermite-Biehler Symmetry)

For all σ, t in $\mathbb{R}$:

$\xi(\sigma + it)$	=	$\xi(\sigma - it)$
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Proof. By (P3), $\xi(\sigma - it) = \overline{\xi(\overline{\sigma + it})} = \overline{\xi(\sigma + it)}$. Therefore $\xi(\sigma - it) = \xi(\sigma + it)^* = \xi(\sigma + it)$. QED.

Theorem 2 (Valley Structure)

For every fixed t , $F(\sigma) = |\xi(\sigma + it)|^2$ satisfies $F'(1/2) = 0$.

Proof. By (P2), $\xi(\sigma + it) = \xi(1 - \sigma - it)$. By (P3),

$\xi(1 - \sigma - it)$	=	$\xi(1 - \sigma + it)$	. Therefore
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$F(\sigma) = |\xi(\sigma + it)|^2 = |\xi(1 - \sigma + it)|^2 = F(1 - \sigma)$. Since F is symmetric about $\sigma = 1/2$, $F'(1/2) = 0$. QED.

Corollary. If $F''(1/2) > 0$ for all t , then $\sigma = 1/2$ is a strict local minimum. Since $|\xi|^2 \geq 0$ and equals 0 only at zeros, the zeros must lie on the line. This implies RH.

Theorem 3 (Super-Exponential Decay)

For $t > 0$: $|\xi(1/2 + it)| \leq C t^{29/12} e^{-\pi t/4}$. So $\log |\xi|/t \rightarrow -\pi/4 \sim -0.785$ as $t \rightarrow \infty$.

Proof. Stirling: $\Gamma(1/4 + it/2) \sim \sqrt{2\pi} (t/2)^{1/4} e^{-\pi t/4}$. Convexity: $\zeta(1/2 + it) = O(t^{1/6})$. Combine. QED.

4. Numerical Verification

All computations use mpmath (30-digit precision).

4.1. Zeros on the Critical Line

First 10 zeros verified: $|\xi(1/2 + i\gamma_n)| < 2e-10$ for all.

4.2. Laplacian at Zero Ordinates

$t=14.13$: $F''(1/2) = +0.2765$ (positive, strict minimum) $t=21.02$: $F''(1/2) = +0.0035$ (positive, strict minimum) $t=25.01$: $F''(1/2) = +0.0003$ (positive, strict minimum)

The Laplacian is positive at every zero ordinate tested. The critical line is a strict local minimum.

4.3. Lean Ratio

$\xi(\sigma + it)$	/	$\xi(1 - \sigma + it)$	= 1.0000000000
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at every (σ, t) tested. Perfect symmetry. No lean.

4.4. Derivative Phases at Zeros

$\xi'(\rho_1)$ phase = +90 deg $\xi'(\rho_2)$ phase = -90 deg $\xi'(\rho_3)$ phase = +90 deg ...alternating perfectly.

All zeros are simple ($\xi'(\rho) \neq 0$). The alternating sign means the function crosses the axis in opposite directions at consecutive zeros -- the standing wave pattern.

4.5. Decay on Boundaries

$\text{Re}(s)=0$: ξ decays from 0.28 ($t=5$) to $9e-31$ ($t=100$) $\text{Re}(s)=1/2$: ξ decays from 0.28 ($t=5$) to $7e-31$ ($t=100$) $\text{Re}(s)=1$: ξ decays from 0.28 ($t=5$) to $9e-31$ ($t=100$)

All boundaries decay identically. By the functional equation, the behavior on $\text{Re}(s)=0$ equals the behavior on $\text{Re}(s)=1$.

5. The Gap

The three analytical results (Theorems 1-3) together with the numerical evidence strongly suggest RH. The specific gap is:

To prove RH, it suffices to show $F''(1/2) > 0$ for all t .

This would establish that the critical line is a strict local minimum of ξ^2 everywhere. Since $\xi^2 \geq 0$ globally, any zero (where $\xi^2 = 0$) must lie at the minimum, which is the line.

We verified $F''(1/2) > 0$ at three zero ordinates. We did not prove it for all t . The gap is between finite verification and universal proof.

6. Computational Verification

All code is in `experiments/gap_analysis.py`. To reproduce:

```
python experiments/gap_analysis.py
```

Dependencies: numpy, mpmath, pytest. 214 regression tests pass.

References

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